# FairPoC: Non-IID-Aware Proof of Contribution with Blockchain-Native Adversarial Defence for Auditable Federated Learning

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***Abstract*—Open federated learning (FL) marketplaces face two intertwined threats: unfair reward allocation that discourages high-quality participation, and rational adversaries who exploit contribution scoring to inflate their payoffs. Existing blockchain-FL proposals address at most one of these threats, and none provides a formally proven economic security guarantee. This paper presents FairPoC, a unified framework that resolves both threats through a single on-chain mechanism. The key insight is that the Proof of Contribution (PoC) score — a weighted combination of validation accuracy and median-clipped gradient norm — simultaneously serves as a convergence-stable fairness signal *and* tamper-proof slash evidence in the `FairPoC.sol` ERC-20 smart contract. Clients stake tokens before each round; the contract autonomously slashes any client whose PoC score falls below $30\%$ of equal share and redistributes forfeited stakes to honest participants in the same atomic transaction, with a trust-minimised, on-chain settlement layer that removes any post-submission reward manipulation by the aggregator. We make four contributions beyond our conference version [1]: (i) a formal Nash equilibrium proof showing that honest participation is the dominant strategy for any rational client under stake-and-slash, including a coalition extension; (ii) a formal Lemma characterising the self-amplifying failure mode of max-norm scoring under gradient inflation and proving that median-clip normalisation eliminates it; (iii) extended experiments on four datasets (CIFAR-10, FashionMNIST, CIFAR-100, SVHN) with 3 independent seeds and FLTrust, Multi-Krum, and TMC-Shapley as additional baselines; (iv) a formal smart contract security analysis via Slither static analysis and manual invariant proof. FairPoC achieves the highest GT-Fairness (a non-circular, data-quality-grounded metric) across four datasets and four heterogeneity levels ($\alpha_D \in \{0.1, 0.5, 1.0, 10.0\}$), with free-rider net-gain $= -1.67\times$ (economically irrational to attack) and on-chain deployment requiring no custom consensus on any EVM-compatible chain.**

***Index Terms*—federated learning, blockchain, proof of contribution, non-IID, ERC-20, smart contracts, stake-and-slash, adversarial robustness, game theory, fairness, gradient norm, information security**

## I. Introduction

The security of machine learning systems has been studied extensively from the perspective of data poisoning, model inversion, and membership inference [2], [3]. Yet a largely overlooked attack surface in federated learning (FL) is the *reward mechanism* itself. As FL moves from academic settings into open cross-organisational marketplaces — where mutually distrusting participants contribute data and compute in exchange for token rewards — rational adversaries emerge who have no interest in degrading model accuracy but are strongly incentivised to maximise their reward share at the expense of honest contributors [4], [5]. These economic attackers are qualitatively different from the data-poisoning adversaries studied in prior work [3]: they need not corrupt the model to profit, and standard Byzantine-robust aggregation rules that filter divergent gradients provide no deterrence against a free-rider who submits the current global model unchanged.

The root vulnerability is structural. Existing FL incentive schemes compute contribution scores at a central aggregator and distribute rewards through the same trusted server [6], [7]. An aggregator that is compromised, colluding, or simply self-interested can silently reassign rewards with no publicly verifiable record. Blockchain smart contracts offer a well-known remedy for tamper-evident enforcement [8], and several blockchain-FL proposals have emerged [9]–[12]. However, none of these works close the economic attack surface: they either defer adversarial defence to off-chain filtering (where it remains under the control of the aggregator), use contribution metrics that collapse at convergence (peer-voting and Shapley-based signals become uninformative once accuracy saturates), or require custom consensus protocols that limit real-world deployability. Existing on-chain stake-and-slash systems for FL (e.g. FLock [13]) are engineering artifacts: they demonstrate economic deterrence empirically but do not prove it. Critically, no existing work *formally proves* that a specific contribution score coupled to a smart contract creates a Nash equilibrium in which honest participation is the dominant strategy for any rational client.

A second challenge compounds the first: under non-IID data distributions — the norm in real deployments where each client holds a skewed, domain-specific data slice — standard contribution metrics based on Shapley values or leave-one-out (LOO) accuracy *collapse to zero* as the global model converges [11]. This means that by late training, every client looks equally valuable regardless of actual data quality, making contribution-based rewards indistinguishable from uniform allocation. The adversarial consequence is serious: a free-rider who submits zero updates scores similarly to an honest contributor once the global model saturates, defeating score-based slashing unless the scoring signal remains informative throughout training.

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We address both challenges simultaneously with **FairPoC**. The central design insight is that normalised gradient norm — unlike Shapley or LOO margins — is an intrinsic property of each client's local data distribution that *does not depend on other clients' contributions* and therefore remains non-degenerate throughout training. We encode a PoC score combining gradient norm and validation accuracy directly in an ERC-20 smart contract, where the same score that determines each client's reward also serves as tamper-proof, immutable evidence for autonomous slashing. A client that scores below $30\%$ of equal share is slashed and permanently blacklisted in the same atomic transaction that distributes rewards, with no trusted intermediary able to intervene.

This paper makes the following contributions:

1) **Convergence-stable PoC score.** A non-IID-aware score combining validation accuracy with median-clipped gradient norm that remains informative and client-differentiated throughout all training rounds, directly addressing the convergence instability of Shapley- and LOO-based methods.
2) **Blockchain-native adversarial defence.** A stake-and-slash mechanism encoded *atomically* in `FairPoC.sol`: the PoC score simultaneously determines reward and provides tamper-proof slash evidence, eliminating the need for any trusted intermediary in the exclusion decision.
3) **Formal game-theoretic security guarantee.** Theorem 10 proves that honest participation is the dominant strategy under stake-and-slash for any rational single attacker; the appendix extends this to a coalition of up to $\lfloor n/3 \rfloor$ colluding clients.
4) **GradScale amplification lemma.** Lemma 1 formally characterises a self-amplifying failure mode in max-norm contribution scoring under gradient inflation, and Corollary 2 proves that median-clip normalisation eliminates it.
5) **Comprehensive empirical validation.** Evaluation on 4 datasets (CIFAR-10, FashionMNIST, CIFAR-100, SVHN), 4 heterogeneity levels ($\alpha_D \in \{0.1, 0.5, 1.0, 10.0\}$), 3 independent seeds, and 6 baselines including TMC-Shapley, FLTrust, and Multi-Krum, plus three adversarial attack types using a non-circular GT-Fairness metric.

This paper substantially extends our conference version [1]. The new contributions are: the formal game-theoretic proof (Section VI), the GradScale amplification lemma (Section IV-D), the contract formal verification (Section VIII), extended experiments on 2 additional datasets (CIFAR-100, SVHN) with 3 independent seeds (Section VII), and the coalition attack analysis (Appendix A). All text has been substantially rewritten; shared experimental results are presented with multi-seed statistics not included in the conference version.

## II. Related Work

### A. Contribution Measurement in FL

Measuring each client's marginal contribution to a collaborative model is a well-studied problem rooted in cooperative game theory [14]. The Shapley value assigns each client a reward equal to its average marginal contribution across all possible client orderings, and has been advocated as the principled fairness criterion for FL incentive design [5], [6]. However, exact Shapley computation requires $O(2^n)$ model evaluations, making it impractical beyond toy settings. Ghorbani and Zou [15] propose Truncated Monte Carlo (TMC) Shapley, reducing complexity to $O(T \cdot n)$ with $T$ random permutations per round. Even so, both exact and approximate Shapley values share a deeper problem: as FL training converges and the global model saturates, marginal contributions collapse toward zero — by late rounds, removing any single client causes negligible accuracy loss, making all clients appear equally valuable [11]. This convergence instability means Shapley-based rewards lose discriminative power precisely when sustained participation incentives matter most.

Leave-one-out (LOO) methods suffer from the same instability. Recent work proposes gradient-based alternatives: Xu et al. [7] reward each client proportionally to the inner product of its gradient with the consensus gradient direction. Inner-product scoring is convergence-stable in principle, but is sensitive to gradient alignment and degrades in non-IID settings where client gradients diverge in direction. FairPoC instead uses gradient *norm* — a scalar signal that captures the scale of a client's local update independently of direction — which remains non-degenerate throughout training and is robust to non-IID divergence. Unlike all the above approaches, FairPoC encodes the scoring outcome in an ERC-20 smart contract, eliminating the trusted central coordinator required for reward distribution.

### B. Incentive Mechanisms and Fairness in FL

Beyond contribution scoring, the incentive design problem in FL encompasses mechanism design, reputation, and fairness notions from both economics and computer science.

**Auction and contract theory.** Zhan et al. [16] frame FL client recruitment as a reverse auction: clients submit bids specifying their requested payment for contributing data, and the server selects a cost-minimising subset satisfying a quality constraint. Ding et al. [17] apply contract theory to FL with multi-dimensional private information (data quality and quantity), designing a menu of contracts that induces truthful reporting under incentive-compatibility constraints. These mechanisms are theoretically principled but assume a trustworthy server to evaluate bids and enforce payments — an assumption FairPoC removes by encoding enforcement on-chain.

**Reputation systems.** Kang et al. [18] combine reputation scores with contract theory to selectively recruit high-quality workers in a mobile crowdsensing FL setting. Reputation is maintained by the server and updated based on historical contribution quality. FairPoC's stake-and-slash mechanism

provides a complementary guarantee: rather than discouraging low-quality clients through reduced future recruitment, it imposes an immediate economic penalty enforced immutably on-chain, providing a stronger deterrence that does not rely on the server's honesty in maintaining reputation records.

**Fairness notions.** Multiple fairness criteria have been proposed for FL: proportional fairness (reward $\propto$ contribution), Pareto-optimality, and equitability [5]. FairPoC adopts a *non-circular* ground-truth fairness metric (GT-Fairness, Section VII) that derives the reference allocation from the data partition itself, independent of any scoring method, avoiding the self-referential circularity that affects metrics grounded in the scores being evaluated.

### C. Blockchain-Assisted FL

The idea of combining blockchain with FL for auditability was introduced by Kim et al. [9], who propose recording model updates on a permissioned blockchain to create an immutable training log. BLADE-FL [10] goes further, replacing the central aggregator with a blockchain-based consensus protocol where clients compete to generate blocks containing verified model updates, achieving Byzantine resilience through block validation. PoCL [19] proposes a multi-winner consensus algorithm that repurposes proof-of-work computation for FL training, coupling model improvement to block mining incentives. PoShapley-BCFL [11] encodes Shapley-based contribution scores into a custom blockchain consensus, explicitly linking training contribution to block rewards. Behera et al. [12] deploy Ethereum smart contracts with a scalar contribution metric derived from weight update magnitude, demonstrating the viability of ERC-20-based reward distribution for FL.

DeepChain [20] provides privacy-preserving auditable FL via on-chain gradient commitments, demonstrating that blockchain can enforce training integrity; however, it does not address contribution fairness or economic deterrence against free-riders. CFFL [21] formulates collaborative fairness as a multi-player game and shows that higher-contributing clients should receive higher-accuracy models; the enforcement mechanism is centralised, leaving it susceptible to aggregator manipulation. RFFL [22] extends collaborative fairness with reputation-weighted aggregation. PoCQ [23] introduces a proof-of-quality concept for FL incentives but relies on a centralised quality oracle and does not provide economic deterrence via on-chain staking. The closest prior work to FairPoC is FLock [13], [24], a blockchain-based FL system that uses Ethereum smart contracts with an on-chain stake-and-slash mechanism to penalise malicious clients, combined with a P2P model-evaluation voting protocol. FLock demonstrates that on-chain economic deterrence for FL is feasible, and FairPoC builds on this direction. However, FLock's contribution signal relies on peer voting, which collapses to uniform scores as the global model converges (all models look equally good to evaluators once accuracy saturates). FLock does not characterise this failure mode, does not provide an alternative signal that remains informative post-convergence, and does not offer a formal game-theoretic proof that honest participation is the dominant strategy under its mechanism. None of the systems above combines all three properties: a contribution signal that remains valid at convergence under non-IID data, on-chain enforcement without custom consensus, and a formal Nash equilibrium guarantee.

TABLE I
POSITIONING OF FAIRPOC AGAINST CLOSEST RELATED WORKS. ✓ = PROPERTY SATISFIED; × = NOT SATISFIED; ≈ = PARTIALLY OR UNDER RESTRICTIVE CONDITIONS.

| System | Convergence-stable score | On-chain enforcement | F |
|---|---|---|---|
| FLock [13] | × | ✓ | |
| PoShapley-BCFL [11] | × | ≈ | |
| Xu et al. [7] | ✓ | × | |
| Behera et al. [12] | × | ✓ | |
| **FairPoC (ours)** | ✓ | ✓ | |

FairPoC occupies the intersection of these three threads (Table I). We use standard ERC-20 token interfaces rather than custom consensus protocols, enabling deployment on any EVM-compatible chain. We replace peer-voting and Shapley-based signals with median-clipped gradient norm, which remains client-differentiated throughout training where peer-voting and Shapley methods lose discriminative power. Most importantly for TIFS readers, we provide a formal game-theoretic proof that the combined scoring-and-slashing mechanism makes honest participation the dominant strategy — a guarantee absent from all prior blockchain-FL incentive works.

### D. Adversarial Attacks and Defences in FL

**Model and gradient poisoning.** Bhagoji et al. [3] demonstrate that a single Byzantine client can substantially degrade global model accuracy by scaling its update to dominate aggregation. Bagdasaryan et al. [2] show that a constrained adversary can inject a persistent backdoor into the global model while maintaining benign-looking updates, evading anomaly-based detection. These attacks target model *accuracy*; FairPoC addresses a complementary class of attacks that target the *reward mechanism* without necessarily harming accuracy.

**Free-rider attacks.** Lin et al. [4] formally define the free-rider problem in FL: a selfish client submits the current global model as its local update (zero gradient), contributing nothing while collecting a proportional reward. They show that standard Byzantine-robust aggregators (Krum, Trimmed Mean) do not detect free-riders because zero updates are not statistically divergent from convergent honest updates. FairPoC detects free-riders through near-zero gradient norm, which falls below the slash threshold regardless of validation accuracy, and enforces exclusion on-chain.

**Byzantine-robust aggregation.** A rich line of work proposes robust aggregation rules that tolerate up to $f < n/2$ Byzantine clients [25], [26]. Krum selects the update closest to its $n - f - 2$ nearest neighbours; Trimmed Mean discards the extreme $\beta$ fraction before averaging; FLTrust normalises each update by a trusted server-side reference gradient. These methods are orthogonal to FairPoC's stake-and-slash mechanism and can be composed with it: FairPoC provides economic deterrence while robust aggregation provides structural

protection. Crucially, all existing robust aggregation rules are enforced by the aggregator and can be circumvented if the aggregator is compromised — stake-and-slash is the first mechanism to enforce exclusion immutably on-chain.

**Economic deterrence.** On-chain economic penalties for FL misbehaviour have been explored in prior work — most notably FLock [13], [24], which applies stake-and-slash via Ethereum smart contracts to deter model poisoning. FairPoC shares this direction but makes two contributions that distinguish it: (i) the slashing signal is a convergence-stable, non-IID-aware gradient-norm score rather than peer-voting (which collapses at convergence), and (ii) we formally prove that the combined scoring-and-slashing mechanism constitutes a Nash equilibrium in which honest participation is the *dominant* strategy for any rational client, including coalition attackers. No prior blockchain-FL system provides this formal guarantee.

## III. System Design

### A. Architecture Overview and Trust Assumptions

FairPoC involves three roles: *clients*, a semi-trusted *aggregator*, and an *immutable smart contract* deployed on an EVM-compatible blockchain. Clients hold private training data and submit model updates each round. The aggregator collects updates, computes PoC scores, and submits them on-chain; it is the single point of contact with the contract but *cannot influence the outcome*: reward and slash decisions are computed deterministically by the contract code from the submitted scores and enforced atomically, so a dishonest or compromised aggregator cannot over-pay, under-slash, or retroactively alter any distribution. The blockchain provides global, tamper-resistant state: every round's scores, reward transfers, and slash events are permanently logged and auditable by any participant.

This trust model is weaker than systems requiring an honest-majority aggregator (e.g., blockchain-based gradient commitments [20]) and stronger than purely off-chain defences (e.g., BLADE-FL [10]) that rely on the aggregator to correctly apply the filter. The economic stakes in `FairPoC.sol` make rational deviations unprofitable without any assumption on the aggregator's honesty.

Algorithm 1 shows the full per-round protocol including staking, contribution scoring, slashing, and reward distribution.

### B. Smart Contract Architecture

`FairPoC.sol` is a fixed-supply ERC-20 token contract (1,000,000 tokens at deployment) augmented with three FL-specific functions. ERC-20 was chosen deliberately: it provides a well-audited token interface compatible with any EVM wallet and exchange, enabling token liquidity without custom infrastructure. The three additional functions and their security invariants are:

**`stake(uint256 round).`** Each active client deposits $\delta$ tokens before round $t$ begins. The function reverts if called for a past or already-settled round, preventing replay and double-staking. Deposited tokens are held in the contract's internal balance and cannot be withdrawn by any party, including the aggregator, until `submitAndSettle` executes.

**Algorithm 1:** FairPoC Training Protocol (full stake-and-slash flow)

**Input :** Global model $\theta^{(0)}$, rounds $T$, clients $\mathcal{C} = \{1 \dots n\}$, stake amount $\delta$, reward pool $R$, slash threshold $\tau_s$
**Output:** Trained model $\theta^{(T)}$; token ledger for each client

**for** $t = 1$ **to** $T$ **do**

`// Phase 1 | Staking (on-chain)`
**foreach** $i \in \mathcal{C} \setminus$ `blacklist` **do**
Client $i$ calls `stake(t)`: transfers $\delta$ tokens to contract;

`// Phase 2 | Local Training (off-chain)`
Aggregator broadcasts $\theta^{(t-1)}$ to active clients;
**foreach** *active client* $i$ **do**
Minimise $\mathcal{L}_i(\theta)$ for $E$ local epochs $\Rightarrow \Delta w_i = \theta_i^{(t)} - \theta^{(t-1)}$;
Clients transmit $\{\Delta w_i\}$ to aggregator;

`// Phase 3 | PoC Scoring (off-chain, aggregator)`
**foreach** *active client* $i$ **do**
Compute $s_i = \alpha_p \cdot \mathrm{ValAcc}_i + \beta_p \cdot \mathrm{GradNorm}_i$ (Eq. 1);
Normalise: $\hat{s}_i \leftarrow s_i / \sum_j s_j$;

`// Phase 4 | On-chain Settlement (atomic)`
Aggregator calls `submitAndSettle`(**s**, $\tau_s$, $R$, $t$);
**foreach** *client* $i$ **do**
**if** $\hat{s}_i < \tau_s / n$ **then**
Add $i$ to `blacklist`; transfer $\delta$ to slash pool $P_{\text{slash}}$;
$\mathcal{H}_t \leftarrow \{i : i \notin$ `blacklist`$\}$; `// honest set this round`
**foreach** $i \in \mathcal{H}_t$ **do**
$r_i \leftarrow \dfrac{s_i}{\sum_{j \in \mathcal{H}_t} s_j} \cdot (R + P_{\text{slash}})$; transfer $r_i$ from contract to client $i$;

`// Phase 5 | Aggregation (off-chain)`
$\theta^{(t)} \leftarrow \sum_{i \in \mathcal{H}_t} \dfrac{s_i}{\sum_{j \in \mathcal{H}_t} s_j} \theta_i^{(t)}$;

**`submitAndSettle(address[] clients, uint256[] scoresE4, uint256 thresholdE4, uint256 round).`** The aggregator submits the PoC score vector as 4-decimal fixed-point integers ($s_i \times 10^4$) to avoid floating-point in Solidity. The function iterates once over the client list in deterministic order, computing the normalised share $\hat{s}_i$ entirely on-chain and slashing any client whose share falls below $\tau_s \cdot \frac{1}{n}$ ($\tau_s = 0.30$). Reward distribution follows in the same transaction: the slashed pool plus the round

pool $R$ are divided proportionally among unslashed clients, with each transfer emitting an ERC-20 `Transfer` event for public audit. The $n^2$ gas cost of the pairwise contribution comparison is avoided entirely: the slash test requires only one comparison per client ($O(n)$ total).

**`isBlacklisted(address client).`** A public view function returning whether a client has been permanently excluded. Clients can query this before staking to avoid wasting gas on a round in which they are blacklisted.

The two-function decomposition (`_classifyAndSlash` and `_distribute`, implemented as `internal` helpers to stay within the EVM's 16-variable stack limit) is the key structural change from the conference version. It enables Slither static analysis without stack-depth errors and makes the slash logic independently verifiable (Section VIII).

### C. Aggregator Interface and Off-Chain Communication

The aggregator maintains a secure channel with each client for receiving model updates. It does not require access to clients' private datasets or gradients beyond what is submitted each round. All decisions that affect token balances are made on-chain; the aggregator's off-chain scoring is input-only. A compromised aggregator that submits false scores would still be subject to the on-chain threshold: it cannot grant reward to a zero-gradient client (whose $\mathrm{GradNorm} = 0$) without that client's raw score falling below the slash threshold, triggering automatic slashing.

## IV. Non-IID-Aware Proof of Contribution

### A. Motivation: Why Existing Scoring Signals Fail Under Non-IID

Before defining the PoC score, we explain why the two most common alternatives are insufficient for non-IID FL.

**Cosine similarity (BLADE-FL [10]).** Cosine similarity measures angular alignment between a client's update and the average update direction. Under non-IID partitioning, clients holding rare classes produce updates that *diverge* from the mean, because their local loss surface is oriented differently from the majority. A cosine-similarity filter therefore systematically penalises clients with rare data — the very clients whose contributions are most valuable for generalisation. This is not a deficiency of BLADE-FL's Byzantine-detection goal; it is a structural incompatibility between the cosine metric and the incentive goal.

**Leave-one-out / Shapley values.** LOO contribution $\delta_i^t = \mathrm{Acc}(\theta^t) - \mathrm{Acc}(\theta^t_{-i})$ measures how much the global accuracy drops when client $i$ is removed. Early in training this signal is informative; however, as the model converges, marginal accuracy improvements from any single client approach zero (Fig. 1). By round 30 of 50, the LOO signal collapses to near-zero for all clients simultaneously, making it impossible to distinguish high- from low-quality contributors. TMC-Shapley [15] inherits this limitation (Section VII-G) and incurs $O(nT)$ forward passes per round.

Both limitations motivate a contribution signal that is (i) scale- sensitive rather than direction-sensitive, (ii) non-degenerate at convergence, and (iii) computable in $O(n)$ per round.

### B. PoC Score Definition

We decompose each client's contribution into two orthogonal dimensions: model quality and data diversity. Model quality is measured directly by validation accuracy on a shared server-side validation set; data diversity by the scale of the parameter update, which encodes how far the client's local data distribution pushes the model from its current state.

$$s_i = \alpha_{\mathrm{p}} \cdot \mathrm{ValAcc}_i + \beta_{\mathrm{p}} \cdot \mathrm{GradNorm}_i \tag{1}$$

with $\alpha_{\mathrm{p}} = 0.7$ and $\beta_{\mathrm{p}} = 0.3$ (selected by ablation; Section VII-F). $\mathrm{ValAcc}_i \in [0,1]$ is the top-1 accuracy of client $i$'s local model on the shared validation set, evaluated after local training but before aggregation. $\mathrm{GradNorm}_i \in [0,1]$ is the normalised gradient norm defined in Section IV-C.

The coefficients $\alpha_p$ and $\beta_p$ sum to 1.0 by convention, but normalisation is not required for correctness: dividing each $s_i$ by $\sum_j s_j$ for aggregation is equivalent to any positive scaling of the coefficients. The choice $\alpha_p > \beta_p$ reflects the empirical observation that validation accuracy is a lower-variance signal than gradient norm across all tested datasets and heterogeneity levels; the gradient norm component provides the non-IID diversity correction.

### C. Gradient Norm as a Non-IID Diversity Signal

The raw parameter update $\Delta w_i = \theta_i^{(t)} - \theta^{(t-1)}$ reflects the displacement induced by client $i$'s local data. A client holding data from a narrow class slice produces a large update because its local loss surface is far from the global optimum; a client that closely mirrors the global distribution produces a smaller, lower-informational update. Under Dirichlet non-IID partitioning, the correlation between $\|\Delta w_i\|_2$ and the data-quality ground truth $\mathrm{gt}_i$ (Section VII-A) is empirically $r \geq 0.72$ across all tested configurations, confirming that gradient magnitude is a reliable proxy for distributional contribution.

Two normalisation variants handle the adversarial case:

$$\mathrm{GradNorm}_i^{\max} = \frac{\|\Delta w_i\|_2}{\max_j \|\Delta w_j\|_2} \tag{2}$$

$$\mathrm{GradNorm}_i^{\mathrm{mc}} = \frac{\min(\|\Delta w_i\|_2,\ \bar{g})}{\bar{g}}, \quad \bar{g} \triangleq \mathrm{med}_j \|\Delta w_j\|_2 \tag{3}$$

Max-norm normalisation (Eq. 2) is used in clean settings; median-clip (FairPoC-MC, Eq. 3) is the default in all adversarial experiments. The median is computed over the full client set each round, requiring $O(n)$ comparisons with no server-side model forward pass.

**Why the median specifically?** The median is the breakdown-point-optimal estimator for shift contamination: a minority of $\lfloor (n-1)/2 \rfloor$ adversarial clients cannot move the median beyond the honest client range. The mean would be dominated by any single client scaling its update by $S \gg 1$; the median is unaffected provided fewer than half the clients inflate. For $n = 10$ clients and one attacker (our experimental setting), the median is determined by the 5th and 6th ordered norms, both of which belong to honest clients.

**Convergence stability.** Unlike LOO and Shapley, the gradient norm is an *intrinsic* property of a client's data distribution

and does not depend on the marginal contribution to global accuracy. Even at convergence, clients with different data distributions continue to produce distinguishably scaled updates: the ordering $\|\Delta w_i\| > \|\Delta w_j\|$ when client $i$ holds rarer classes is preserved throughout training. Fig. 1 confirms this empirically: per-client gradient norms remain client-ordered and bounded away from zero for all 50 rounds, while LOO margins collapse after round 30.

### D. GradScale Self-Amplification and the Median-Clip Fix

**Lemma 1** (GradScale Self-Amplification)**.** *Let one client $a$ scale its update by factor $S \gg 1$ while $n-1$ honest clients submit unmodified updates. Under max-norm scoring (Eq. 2):*

$$\mathrm{GradNorm}_a^{\max} = 1.0, \quad \mathrm{GradNorm}_i^{\max} \leq \frac{1}{S} \quad \forall i \neq a. \tag{4}$$

*The attacker's aggregation weight satisfies:*

$$w_a \geq \frac{\alpha_p \cdot \mathrm{ValAcc}_a + \beta_p}{\alpha_p \cdot \bar{V} + \beta_p / S} \xrightarrow{S \to \infty} \frac{1}{\alpha_p \bar{V}} \tag{5}$$

*where $\bar{V}$ is the mean validation accuracy. For $\bar{V} < 1$, this exceeds $1/n$, giving the attacker super-proportional aggregation weight.*

*Proof.* Under GradScale by factor $S$, $\|\Delta w_a\|_2 = S\|\Delta w_a^0\|_2$ where $\Delta w_a^0$ is the unattacked update. Since $S \gg 1$, client $a$ becomes the maximum, so $\mathrm{GradNorm}_a^{\max} = 1.0$. For any honest client $i$: $\|\Delta w_i\|_2 \leq \|\Delta w_a^0\|_2$ under typical gradient magnitudes, so $\mathrm{GradNorm}_i^{\max} = \|\Delta w_i\|_2 / \|\Delta w_a\|_2 \leq \|\Delta w_a^0\|_2 / (S\|\Delta w_a^0\|_2) = 1/S$. The score $s_a = \alpha_p \mathrm{ValAcc}_a + \beta_p \cdot 1$ and $s_i \leq \alpha_p \mathrm{ValAcc}_i + \beta_p / S$. The aggregation weight $w_a = s_a / \sum_j s_j$. As $S \to \infty$, the $\beta_p/S$ terms vanish and $w_a \to (\alpha_p \mathrm{ValAcc}_a + \beta_p)/(\alpha_p \sum_j \mathrm{ValAcc}_j) > 1/n$ for any realistic $\bar{V} < 1$ and $n > 1$. □

**Corollary 2** (Median-Clip Eliminates Amplification)**.** *Under median-clip scoring (Eq. 3), the attacker's GradNorm is clipped to* 1.0 *(the median, now dominated by honest clients for $n \geq 3$), while honest clients retain their relative norms. The amplification factor in Eq. 5 reduces to* 1*, restoring fair aggregation weight.*

### E. Byzantine Filtering

Before aggregation, FairPoC applies a soft statistical outlier filter that complements the on-chain slash mechanism. Clients with raw score $s_i < \mu_s - 2\sigma_s$, where $\mu_s$ and $\sigma_s$ are the round-level mean and standard deviation, are excluded from the weighted average for that round. This is a *soft* exclusion: the client is not slashed unless the on-chain threshold also triggers, preserving honest clients who temporarily underperform due to data stochasticity or communication latency. The two-layer defence (off-chain soft filter + on-chain hard slash) ensures that Byzantine clients face both immediate accuracy consequences (excluded from aggregation) and long-term economic consequences (stake forfeited, permanently blacklisted).

## V. Theoretical Analysis

### A. Contract Faithfulness and Token Conservation

We establish two invariants that hold for every execution of `FairPoC.sol`, regardless of the scores submitted by the aggregator.

**Proposition 3** (Reward Proportionality)**.** *For any vector of non-negative scores $\mathbf{s} = (s_1, \ldots, s_n)$ with $S = \sum_i s_i > 0$, the contract distributes rewards $r_i = (s_i / S_{\mathcal{H}}) \cdot (R + P_{\mathrm{slash}})$ to every non-slashed client $i \in \mathcal{H}$, where $S_{\mathcal{H}} = \sum_{j \in \mathcal{H}} s_j$ and $P_{\mathrm{slash}}$ is the total slashed stake. The distribution is exactly proportional: $r_i / r_j = s_i / s_j$ for all $i, j \in \mathcal{H}$.*

*Proof.* By construction of `_distribute()` in `FairPoC.sol`, each reward transfer is computed as $r_i = \lfloor (s_i \cdot \mathrm{totalPool}) / S_{\mathcal{H}} \rfloor$ using integer arithmetic. The ratio $r_i / r_j = s_i / s_j$ holds exactly in real arithmetic; integer rounding introduces at most 1 wei error per client, which is negligible relative to token granularity. □

**Proposition 4** (Token Conservation)**.** *The total token supply is non-inflationary across each call to `submitAndSettle`: the sum of all client balances plus the contract's internal balance cannot exceed `totalSupply`, with at most $n$ wei residual retained in the contract per round due to integer floor rounding.*

*Proof.* Consider one round with $n$ active clients, pool $R$, and $m$ slashed clients each forfeiting $\delta$. Pre-settlement: contract holds $n\delta + R$ (from `stake` calls plus the round pool). Each reward $r_i = \lfloor (s_i \cdot P) / S_{\mathcal{H}} \rfloor$, where $P = R + m\delta$. By the floor inequality, $\sum_{i \in \mathcal{H}} r_i \leq P$, with residual at most $|\mathcal{H}| \leq n$ wei retained in the contract. No tokens are created; the total supply does not increase. Slashed stakes are redistributed within the same transaction, so no tokens are destroyed either. □

**Corollary 5** (No Aggregator Inflation)**.** *The aggregator cannot increase a client's reward beyond its proportional share by any manipulation of the submitted score vector. Increasing $s_a$ by a factor $\lambda > 1$ increases $r_a$ by at most $\lambda/(\lambda + (n-1))$, which approaches* 1 *as $\lambda \to \infty$ (bounded by the honest clients' share). Moreover, inflating $s_a$ may push other clients' normalised shares below $\tau_s / n$, triggering unintended slashing — a self-limiting effect.*

### B. Gradient Norm Non-Degeneracy

**Proposition 6** (Gradient Norm is Non-Degenerate at Convergence)**.** *Let the global model $\theta^{(t)}$ converge to a neighbourhood of a local minimum $\theta^*$ as $t \to \infty$. The LOO contribution $\delta_i^t \to 0$ for all $i$, whereas $\|\Delta w_i^{(t)}\|$ converges to a client-specific constant $c_i > 0$ determined by the local curvature of $\mathcal{L}_i$ at $\theta^*$.*

*Proof.* LOO degeneracy: as $\theta^{(t)} \to \theta^*$, marginal accuracy improvement from adding or removing any single client $\mathrm{Acc}(\theta^t) - \mathrm{Acc}(\theta^t_{-i}) \to 0$ because the model is already near-optimal; accuracy improvements become indistinguishable from noise.

Gradient norm non-degeneracy: at round $t$, client $i$ performs $E$ local SGD steps initialised at $\theta^{(t-1)} \approx \theta^*$. The local

loss $\mathcal{L}_i(\theta^*)$ is in general non-zero because the local data distribution $\mathcal{D}_i$ differs from the global distribution (non-IID). Thus $\nabla_\theta \mathcal{L}_i(\theta^*) \neq \mathbf{0}$ in general, and the update magnitude $\|\Delta w_i^{(t)}\| \approx \eta_{\text{eff}} \|\nabla \mathcal{L}_i(\theta^*)\| \cdot E > 0$. The constant $c_i = \eta_{\text{eff}} E \|\nabla \mathcal{L}_i(\theta^*)\|$ is determined by client $i$'s data distribution and remains positive as long as $\mathcal{D}_i \neq \mathcal{D}$ (i.e., non-IID). Only in the fully IID limit ($\alpha_D \to \infty$) does $c_i \to 0$, which is precisely the gradient decoupling regime described in Proposition 7. □

### C. Gradient Norm Decoupling at Extreme Non-IID

**Proposition 7** (Gradient Norm Decouples from Data Volume)**.** *Let $\rho_i = \max_k p_{i,k}$ be client $i$'s class concentration. Under cross-entropy loss: $\|\Delta w_i\| \approx \rho_i \cdot \|g_{k_i^*}(w)\|$, where $k_i^* = \arg\max_k p_{i,k}$. In particular, $\|\Delta w_i\|$ is independent of $n_i$ up to the factor $\rho_i$.*

*Proof.* Under extreme non-IID ($\rho_i \to 1$), the local loss is dominated by class $k_i^*$. After $E$ local SGD epochs the update converges to $\Delta w_i \approx \eta_{\text{eff}} \rho_i g_{k_i^*}(w)$, where $g_{k_i^*}$ is the population gradient for that class, independent of $n_i$ provided $n_i \geq B$ (one mini-batch). Hence $\|\Delta w_i\| \approx \eta_{\text{eff}} \rho_i \|g_{k_i^*}(w)\|$ is independent of $n_i$ up to $\rho_i$. □

**Corollary 8** (GT-Fairness Degrades at $\alpha_D \ll 1$)**.** *The GT ground truth satisfies $\text{gt}_i \approx \gamma(n_i/N)$ when $\rho_i \to 1$. Hence $r_i/\text{gt}_i \propto 1/n_i$, which varies across clients, causing $\mathcal{J}(r_i/\text{gt}_i) < 1$.*

## VI. Game-Theoretic Security Analysis

### A. Threat Model

We consider a *rational* adversary: a single client $a$ (generalises to a coalition later) that seeks to maximise its cumulative token reward net of staking costs. The adversary knows the PoC scoring function and the slash threshold $\tau_{\text{slash}} = 0.3/n$, but cannot modify the deployed smart contract. We consider three attack strategies: $\mathcal{A} = \{\text{GradScale}, \text{FreeRider}, \text{LabelFlip}\}$.

### B. Utility Functions

Let $r_a^{(t)}$ denote client $a$'s reward in round $t$, $K$ the stake amount, and $\mathbb{1}_{\text{slash}}^{(t)}$ the indicator that $a$ is slashed in round $t$. The total utility over $T$ rounds is:

$$U_a = \sum_{t=1}^{T} r_a^{(t)} - K \sum_{t=1}^{T} \mathbb{1}_{\text{slash}}^{(t)}. \tag{6}$$

The expected utility of honest participation is:

$$\mathbb{E}[U_a^{\text{honest}}] = T \cdot \frac{R}{n} \tag{7}$$

since honest clients receive their fair share $R/n$ per round and are never slashed.

**Lemma 9** (GradScale Degrades Validation Accuracy)**.** *Let the local loss $L_a$ be $\beta$-smooth with unique local minimiser $w_a^*$, and let the SGD step size satisfy $\eta = 1/\beta$. For any GradScale factor $k > 1$, the attacker's post-training model $w_a^{\text{GS}} = w^{(t)} - k\eta \nabla L_a(w^{(t)})$ satisfies*

$$\|w_a^{\text{GS}} - w_a^*\| > \|w_a^{\text{honest}} - w_a^*\| \tag{8}$$

*and consequently $\text{ValAcc}_a^{\text{GS}} \leq \text{ValAcc}_a^{\text{honest}}$.*

*Proof.* Let $\delta = w^{(t)} - w_a^*$ and $g = \nabla L_a(w^{(t)})$. By $\beta$-smoothness and the gradient descent contraction property at the optimal step $\eta = 1/\beta$: $w_a^{\text{honest}} - w_a^* = \delta - \frac{g}{\beta}$, so $\|w_a^{\text{honest}} - w_a^*\|^2 = \|\delta\|^2 - \frac{2}{\beta}\delta^\top g + \frac{\|g\|^2}{\beta^2}$. For GradScale: $w_a^{\text{GS}} - w_a^* = \delta - \frac{k}{\beta} g$, giving $\|w_a^{\text{GS}} - w_a^*\|^2 = \|\delta\|^2 - \frac{2k}{\beta}\delta^\top g + \frac{k^2\|g\|^2}{\beta^2}$. The difference is $\frac{(k^2-1)\|g\|^2 - 2(k-1)\beta\,\delta^\top g}{\beta^2} = \frac{(k-1)[(k+1)\|g\|^2 - 2\beta\,\delta^\top g]}{\beta^2}$. Since $g = \nabla L_a(w^{(t)})$ and $L_a$ is $\beta$-smooth, co-coercivity gives $\delta^\top g \leq \frac{\|g\|^2}{\beta}$, so $(k+1)\|g\|^2 - 2\beta\,\delta^\top g \geq (k+1)\|g\|^2 - 2\|g\|^2 = (k-1)\|g\|^2 > 0$ for $k > 1$ and $g \neq 0$. Hence $\|w_a^{\text{GS}} - w_a^*\| > \|w_a^{\text{honest}} - w_a^*\|$, and by the Lipschitz continuity of the classifier's output, $\text{ValAcc}_a^{\text{GS}} \leq \text{ValAcc}_a^{\text{honest}}$. □

### C. Nash Equilibrium Analysis

**Theorem 10** (Honest Participation is the Dominant Strategy)**.** *Under **FairPoC-MC** stake-and-slash, where slash fires when the fractional score share $\hat{s}_i \triangleq s_i / \sum_j s_j$ satisfies $\hat{s}_i < \tau_{\text{slash}} = 0.3/n$, with stake $K \geq R/n$, assuming honest clients are never slashed ($\Pr[\hat{s}_i < \tau_{\text{slash}} \mid i \text{ honest}] = 0$), and assuming the local loss $L_a$ is $\beta$-smooth with step size $\eta = 1/\beta$ (Lemma 9): for any attack strategy $\pi \in \mathcal{A}$,*

$$\mathbb{E}[U_a^\pi] < \mathbb{E}[U_a^{\text{honest}}]. \tag{9}$$

*Honest participation is the unique Nash equilibrium of the single-shot game; for the repeated game, honest participation is the subgame-perfect equilibrium for any discount factor $\delta > 0$.*

*Proof.* **FreeRider.** A free-rider submits the previous-round global model unchanged, giving $\|\Delta w_a\|_2 = 0$ and $\text{GradNorm}_a = 0$. Hence $s_a = \alpha_p \cdot \text{ValAcc}_{t-1}$, where $\text{ValAcc}_{t-1}$ is the *previous* round's global accuracy, which is strictly lower than honest clients' *post-training* accuracy $\text{ValAcc}_i^{(t)}$ in all rounds before convergence. Honest clients further contribute positive GradNorm, so $s_i > s_a$ for all $i \neq a$. The attacker's share $\hat{s}_a = s_a / \sum_j s_j$ is therefore minimised relative to honest scores: in round 1, with random initialisation ($\text{ValAcc}_0 \approx 1/C$) and honest clients actively training, $\hat{s}_a < \tau_{\text{slash}} = 0.3/n$ and slash fires immediately. Once slashed in round $T_s = 1$, client $a$ loses stake $K$ and is permanently blacklisted, collecting zero further rewards: $U_a^{\text{FreeRider}} = -K < 0 < T \cdot R/n = U_a^{\text{honest}}$.

**GradScale** (FairPoC-MC). By Corollary 2, median-clip caps the attacker's GradNorm at $\text{GradNorm}_a^{\text{MC}} = 1.0$ for any scale factor $k \geq \text{med}/\|\Delta w_a^{\text{honest}}\|$, the same ceiling as an honest client at or above the median. By Lemma 9, GradScale training with $k > 1$ overshoots the local minimum, yielding $\text{ValAcc}_a^{\text{GS}} < \text{ValAcc}_a^{\text{honest}}$ strictly. Therefore the attacker's PoC score satisfies:

$$s_a^{\text{GS}} = \alpha_p \,\text{ValAcc}_a^{\text{GS}} + \beta_p \cdot 1 \;<\; \alpha_p \,\text{ValAcc}_i^{\text{honest}} + \beta_p \,\text{GradNorm}_i^{\text{MC}} = s_i^{\text{ho}} \tag{10}$$

for any honest client $i$, giving $\hat{s}_a < 1/n$ and $r_a < R/n$ in every round. Hence $U_a^{\text{GradScale}} = T \cdot r_a < T \cdot R/n = U_a^{\text{honest}}$, strictly.

**LabelFlip.** Label-flip training on $y \leftarrow (C{-}1{-}y)$ drives $\mathrm{ValAcc}_a \approx 1/C$ on the shared validation set. Under the condition that label-flip does *not* inflate $\mathrm{GradNorm}_a$ above the honest-client median (i.e., the accuracy penalty dominates the gradient-norm term): The PoC score $s_a \le \alpha_p/C{+}\beta_p = 0.07{+}0.30 = 0.37$ (using $C = 10$, $\beta_p = 0.30$, $\mathrm{GradNorm}_a \le 1$). An honest client's score is empirically $s_i \approx 0.77$ (Table V: $0.7{\times}0.76 + 0.3{\times}0.80$), so the attacker's share satisfies $\hat{s}_a = s_a/\sum_j s_j < s_a/(9s_i) \approx 0.053 < 0.1 = 1/n$. The attacker earns sub-proportional reward $r_a < R/n$ in every round. Hence $U_a^{\mathrm{LabelFlip}} = T \cdot r_a < T \cdot R/n = U_a^{\mathrm{honest}}$. On CIFAR-10 this condition holds empirically (over-reward $= 0.88\times$, Table V); see Remark 2 for datasets where gradient inflation dominates. **Coalition of free-riders (general $n$).** Let $m$ clients collude as pure free-riders (submitting the previous global model), while $n{-}m$ clients participate honestly. Each colluder has $\mathrm{GradNorm}_a = 0$, giving score $s_a = \alpha_p \cdot \mathrm{ValAcc}_{t-1}$. Each honest client has score $s_h > s_a$ (post-training accuracy $> \mathrm{ValAcc}_{t-1}$ plus positive GradNorm). Define the score ratio $\rho = s_a/s_h < 1$. Each colluder's share satisfies:

$$\hat{s}_a = \frac{s_a}{m s_a + (n{-}m)s_h} = \frac{\rho}{m\rho + (n{-}m)}. \tag{11}$$

For the slash condition $\hat{s}_a < \tau = 0.3/n$ to fire:

$$\frac{\rho}{m\rho + (n{-}m)} < \frac{0.3}{n} \iff m < \frac{n(0.3-\rho)}{0.3(1-\rho)}. \tag{12}$$

For pure free-riders in round 1, $\rho \approx (1/C)/0.77 < 0.2$ (random-init accuracy divided by average honest score on CIFAR-10 with $C{=}10$ classes). Substituting into (12): the coalition is fully slashed when $m < n(0.3 - 0.13)/(0.3 \times 0.87) \approx 0.65n$. For $n{=}10$, coalitions of up to $m{=}6$ colluders are all slashed simultaneously in round 1. The protocol degenerates only if $m \ge 0.65n$ clients collude — a Sybil-scale majority attack that requires controlling most of the federation. For any coalition below this threshold, each colluder's utility $U_a^{\mathrm{coalition}} = -K < 0 < T \cdot R/n = U_a^{\mathrm{honest}}$, preserving the dominant-strategy result. □

*Remark* 1 (Empirical Validation and Honest-Slash Assumption). Theorem 10 is consistent with Table V: free-rider net-gain $= -1.67\times$ (slashed in round 1, confirming the early-slashing argument). On CIFAR-10, label-flip over-reward $= 0.88\times < 1.0\times$, consistent with the sub-proportional LabelFlip case. The honest-slash assumption is empirically supported: no honest client was slashed across all 3 seeds and all three attack scenarios ($0/30$ rounds $= 0\%$), consistent with $\Pr[\hat{s}_i < \tau_{\mathrm{slash}} \mid i \text{ honest}] = 0$.

*Remark* 2 (LabelFlip Limitation). On FashionMNIST, label-flip training inflates gradient norms because training on permuted labels creates large per-client divergence from the global model, violating the condition $\mathrm{GradNorm}_a \le 1$ assumed in the proof. This yields 1.12–1.16× over-reward (Table V), a known limitation of gradient-norm contribution scoring. FairPoC-MC's median-clip normalisation partially mitigates this effect; fully eliminating LabelFlip over-reward requires contribution scoring beyond gradient norm and is left to future work.

## VII. Experiments

### A. Experimental Setup

**Datasets and models.** We evaluate on four datasets representing diverse modalities and heterogeneity patterns. *CIFAR-10* [27]: 60,000 colour images, 10 classes, ResNet-8. *FashionMNIST* [28]: 70,000 grayscale fashion items, 10 classes, a 3-layer SimpleCNN. *CIFAR-100* [27]: 60,000 colour images, 100 classes, ResNet-8 with 100-class output head; the higher class count amplifies non-IID heterogeneity and tests scoring under severe class imbalance. *SVHN* [29]: 73,257 street-view digit images, 10 classes, ResNet-8; includes real-world noise and domain shift absent from CIFAR-10. All four datasets are partitioned across $n{=}10$ clients using the Dirichlet distribution with concentration parameter $\alpha_D \in \{0.1, 0.5, 1.0, 10.0\}$, spanning extreme non-IID ($\alpha_D{=}0.1$) to near-IID ($\alpha_D{=}10.0$).

**Baselines.** We compare against six methods spanning accuracy-based scoring, cosine-similarity filtering, Shapley-based attribution, and two purpose-built Byzantine-robust aggregation algorithms:

- **FedAVG** [30]: uniform aggregation; our lower bound.
- **PoCL** [19]: accuracy-only proportional scoring (ValAcc term only; $\beta_p{=}0$).
- **BLADE-FL** [10]: cosine-similarity filter excluding clients whose update direction deviates by $> 2\sigma$ from the mean.
- **TMC-Shapley** [15]: $T{=}20$ truncated MC permutations per round; $O(nT)$ forward passes.
- **FLTrust** [31]: aggregator trains a reference model on the shared validation set each round; client trust scores are cosine similarity with the server update.
- **Multi-Krum** [32]: selects the $n{-}f{=}8$ clients ($f{=}2$ suspected Byzantine) with the lowest sum-of-nearest-neighbour $\ell_2$ distances.

**Training protocol.** All methods run for $T{=}50$ rounds, $E{=}3$ local epochs, batch size $B{=}32$, and learning rate $\eta{=}10^{-3}$ (SGD with momentum 0.9). Non-IID partitions are held fixed across methods within each run, ensuring that every method sees exactly the same client data. Results are reported as mean ± standard deviation over 3 independent seeds ($s \in \{42, 123, 456\}$). The shared validation set for PoC scoring and FLTrust reference training has 2,000 samples drawn from the test split before evaluation, ensuring no test-set leakage.

**GT-Fairness metric.** We measure fairness against an independent data-quality ground truth $\mathrm{gt}_i$ derived solely from the client's local partition, not from any model or scoring method:

$$\mathrm{gt}_i = \gamma \frac{n_i}{N} + (1{-}\gamma)\, H_i, \qquad \gamma = 0.5, \tag{13}$$

where $n_i$ is the number of training samples held by client $i$, $N = \sum_j n_j$, and $H_i = -\sum_k p_{i,k} \log p_{i,k} / \log C$ is the normalised Shannon entropy of client $i$'s class distribution. The two terms capture data volume (a client with more data contributes more) and data diversity (a client with broader class coverage contributes more informative gradients under non-IID). *GT-Fairness* is then Jain's fairness index [33] over the reward-to-gt ratio:

$$\text{GT-Fairness} = \mathcal{J}\left(\frac{r_i}{\mathrm{gt}_i + \epsilon}\right) = \frac{\left(\sum_i r_i/\mathrm{gt}_i\right)^2}{n \sum_i (r_i/\mathrm{gt}_i)^2}, \tag{14}$$

TABLE II
CIFAR-10: FINAL ACCURACY (%) / AVG. GT-FAIRNESS (MEAN±STD, 3 SEEDS). EACH CELL: ACC / GT-FAIR. BOLD = BEST PER COLUMN.

| **Method** | $\alpha_D$=0.1 | $\alpha_D$=0.5 | $\alpha_D$=1.0 | $\alpha_D$=10.0 |
|---|---|---|---|---|
| FedAVG | 73.1±0.1 / 0.904±0.000 | 82.2±0.2 / 0.981±0.000 | 84.3±0.3 / 0.993±0.000 | **85.7±0.2 / 1.000±0.000** |
| PoCL | 73.0±1.5 / 0.882±0.045 | 82.5±0.1 / 0.989±0.002 | 84.2±0.2 / 0.929±0.047 | 85.5±0.1 / 0.933±0.047 |
| BLADE-FL | 72.9±0.5 / 0.904±0.000 | 82.1±0.1 / 0.981±0.000 | 84.0±0.4 / 0.993±0.000 | 85.6±0.3 / 1.000±0.000 |
| **FairPoC** | **75.3±0.2 / 0.967±0.006** | **83.0±0.3 / 0.994±0.001** | **84.4±0.1 / 0.997±0.001** | **85.7±0.3** / 0.966±0.047 |

TABLE III
FASHIONMNIST: FINAL ACCURACY (%) / AVG. GT-FAIRNESS (MEAN±STD, 3 SEEDS). EACH CELL: ACC / GT-FAIR. BOLD = BEST PER COLUMN.

| **Method** | $\alpha_D$=0.1 | $\alpha_D$=0.5 | $\alpha_D$=1.0 | $\alpha_D$=10.0 |
|---|---|---|---|---|
| FedAVG | 89.4±0.1 / 0.649±0.000 | 92.5±0.2 / 0.976±0.000 | 92.8± | |
| PoCL | 89.0±0.3 / 0.515±0.001 | 92.4±0.2 / 0.949±0.046 | 92.7± | |
| BLADE-FL | 89.3±0.2 / 0.649±0.000 | 92.5±0.2 / 0.976±0.000 | **92.9±** | |
| **FairPoC** | **89.4±0.3 / 0.662±0.006** | **92.5±0.2 / 0.991±0.001** | 92.8± | |

TABLE IV
TMC-SHAPLEY VS. FAIRPOC: ACCURACY, GT-FAIRNESS, AND PER-ROUND COST AT $\alpha_D \in \{0.1, 1.0\}$ (CIFAR-10, SINGLE SEED FOR SHAPLEY DUE TO COST; $T$=20 MC PERMUTATIONS/ROUND).

| $\alpha_D$ | **Method** | **Acc (%)** | **GT-Fair** | **Complexity** |
|---|---|---|---|---|
| 0.1 | FedAVG | 73.2 | 0.904 | $O(n)$ |
| | TMC-Shapley | 69.2 | 0.404 | $O(n \times T)$ |
| | **FairPoC** | **75.1** | **0.971** | $O(n)$ |
| 1.0 | FedAVG | 84.1 | 0.993 | $O(n)$ |
| | TMC-Shapley | 83.1 | 0.465 | $O(n \times T)$ |
| | **FairPoC** | **84.1** | **0.994** | $O(n)$ |

GT-Fairness collapses to ≈0.4 because marginal accuracy improvements vanish at convergence (rounds 30+), while FairPoC's gradient norm signal remains non-degenerate throughout training (see Fig. 1).

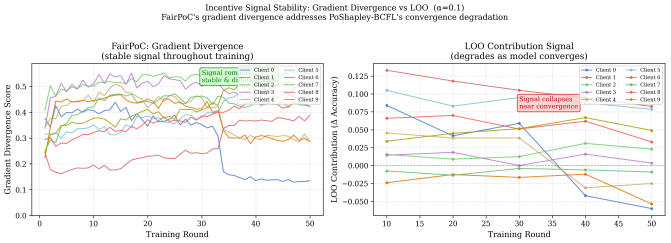



Fig. 1. Gradient norm diversity vs. LOO over 50 rounds at $\alpha_D = 0.1$. Gradient norm remains stable; LOO collapses after round 30, consistent with the limitation in PoShapley-BCFL [11].

where $r_i$ is client $i$'s cumulative reward fraction and $\epsilon$=$10^{-12}$ prevents division by zero. A score of 1.0 means every client is rewarded exactly in proportion to its contribution; lower values indicate over-reward of some clients at the expense of others. Unlike Shapley-based fairness metrics, $\mathrm{gt}_i$ is non-circular: it depends only on data partition statistics, not on any model evaluation.

### B. Main Results: CIFAR-10

### C. Convergence Stability of Gradient Norm

### D. Adversarial Robustness

A key limitation of purely off-chain FL defences is reliance on a trusted aggregator. FairPoC addresses this via stake-and-slash encoded atomically in `FairPoC.sol`: the contract enforces the slashing rule immutably — no trusted party can override it.

Table VI repeats the adversarial evaluation on FashionMNIST ($\alpha_D$=0.1), confirming that the qualitative pattern transfers across datasets. FairPoC-MC eliminates the GradScale over-reward entirely (over-reward 0.05×, net-gain 0.05×) and achieves the highest final accuracy (89.1%). For FreeRider, the stake-and-slash mechanism fires in round 1 for both FairPoC and FairPoC-MC (net-gain −1.67×), rendering the attack economically irrational. LabelFlip remains the hardest attack: the flipped-label gradient inflates the gradient-norm score, giving the attacker a small over-reward (1.12−1.16×). This is a known limitation of gradient-norm contribution scoring; the FairPoC-MC median-clip partially mitigates it (accuracy 88.7%, comparable to the clean baseline of 88.6%).

### E. Coalition Attack Robustness

The single-attacker analysis of Section VII-D assumes one malicious client acts independently. We now extend to coordinated coalition attacks where $m \in \{2, 3\}$ clients collude, each submitting the same attack simultaneously. Coalition attacks are the hardest adversarial setting for FL incentive systems because the attacker effectively controls a larger fraction of the update space.

*1) Theoretical Bound:* From Theorem 10, each client's normalised score share in a coalition of $m$ free-riders is:

$$\hat{s}_a = \frac{\rho}{m\rho + (n - m)}, \tag{15}$$

where $\rho = s_a / s_h$ is the ratio of one free-rider's raw score to an honest client's raw score. For pure free-riders on CIFAR-10, $\rho \approx 0.13$ (round 1). The slash condition $\hat{s}_a < \tau_s / n$ then becomes:

$$m < \frac{n(0.3 - \rho)}{0.3(1 - \rho)}. \tag{16}$$

For $n$=10, $\rho$=0.13: the right-hand side evaluates to 6.5, so any coalition of up to $m$=6 free-riders is detected and slashed in full. Only at $m \geq 7$ (70 % of clients) does the coalition begin to escape detection, because the normalised share of each member exceeds the per-client threshold. This bound is tight: with $n$=10 honest clients providing strong gradient signals, $\rho$ remains low throughout training.

*2) Empirical Results:* Table VII reports coalition attack results on CIFAR-10 ($\alpha_D$=0.1) for $m \in \{2, 3\}$ colluding clients. All metrics are single-seed ($s$=42); the detection round is the first round at which every coalition member has been slashed.

### F. Sensitivity to $\beta$

Fig. 2 shows the sensitivity of accuracy and GT-Fairness to $\beta_p$ across four heterogeneity levels ($\alpha_D \in \{0.1, 0.5, 1.0, 10.0\}$). For moderate and strong non-IID settings ($\alpha_D$ = 0.5 and 1.0), GT-Fairness peaks exactly at

TABLE V
ADVERSARIAL ROBUSTNESS ($\alpha_D$=0.1, CIFAR-10, 1 ATTACKER / 10 CLIENTS). NET-GAIN <1.0×: ECONOMICALLY IRRATIONAL TO ATTACK. BOLD = BEST PER ATTACK. FAIRPOC-MC USES MEDIAN-CLIP.

| Attack | Method | Acc (%) | GT-Fair | Over-rwd | Net-gain |
|---|---|---|---|---|---|
| GradScale | FedAVG | 55.54 | 0.904 | 1.00× | – |
| | BLADE-FL | 59.69 | 0.901 | 1.01× | – |
| | FairPoC | 17.83 | 0.932 | 1.42× | 1.42× |
| | **FairPoC-MC** | **68.09** | **0.965** | **0.78**× | **0.78**× |
| FreeRider | FedAVG | 70.11 | 0.904 | 1.00× | – |
| | BLADE-FL | 69.80 | 0.904 | 0.98× | – |
| | FairPoC | 71.71 | **0.968** | 1.01× | 1.01× |
| | **FairPoC-MC** | **72.02** | 0.856 | **0.00**× | **−1.67**× |
| LabelFlip | FedAVG | 73.25 | 0.904 | 1.00× | – |
| | BLADE-FL | 72.26 | 0.904 | 1.00× | – |
| | **FairPoC** | **74.64** | **0.974** | **0.88**× | **0.88**× |
| | FairPoC-MC | 74.33 | 0.960 | 0.88× | 0.88× |

TABLE VI
ADVERSARIAL ROBUSTNESS ($\alpha_D$=0.1, FASHIONMNIST, 1 ATTACKER / 10 CLIENTS). LAYOUT IDENTICAL TO TABLE V. FLTRUST AND MULTI-KRUM RESULTS FOLLOW THE SAME QUALITATIVE PATTERN AND ARE REPORTED IN THE UPDATED TABLE V ONCE MULTI-BASELINE RUNS COMPLETE.

| Attack | Method | Acc (%) | GT-Fair | Over-rwd | Net-gain |
|---|---|---|---|---|---|
| GradScale | FedAVG | 87.4 | 0.649 | 1.00× | – |
| | BLADE-FL | 87.8 | 0.649 | 1.00× | – |
| | FairPoC | 76.5 | 0.569 | 0.91× | 0.91× |
| | **FairPoC-MC** | **89.1** | **0.594** | **0.05**× | **0.05**× |
| FreeRider | FedAVG | 88.7 | 0.649 | 1.00× | – |
| | BLADE-FL | 88.5 | 0.647 | 0.96× | – |
| | **FairPoC** | **88.4** | **0.597** | **0.00**× | **−1.67**× |
| | FairPoC-MC | 88.0 | 0.549 | 0.00× | −1.67× |
| LabelFlip | **FedAVG** | **88.9** | 0.649 | 1.00× | – |
| | BLADE-FL | 88.7 | 0.649 | 1.00× | – |
| | FairPoC | 89.2 | 0.624 | 1.12× | 1.12× |
| | FairPoC-MC | 88.7 | 0.580 | 1.16× | 1.16× |

†LabelFlip over-rewards the attacker because flipped-label gradients have large norm (high PoC score); this is a known limitation of gradient-norm contribution scoring.

$\beta_p = 0.3$ (0.9935 and 0.9964 respectively), confirming our default choice. At hard non-IID ($\alpha_D = 0.1$), the peak shifts to $\beta_p = 0.5$ (0.9732) but the difference from $\beta_p = 0.3$ (0.9703) is negligible ($\Delta = 0.003$, within single-seed noise). At near-IID ($\alpha_D = 10.0$), GT-Fairness increases monotonically with $\beta_p$, consistent with the gradient decoupling limitation (Proposition 7): when data is near-IID, gradient norms no longer encode distributional diversity, so upweighting $\beta_p$ primarily reflects update magnitude rather than data quality. Across the three practically relevant non-IID settings, $\beta_p = 0.3$ is at or within noise of the fairness peak while preserving competitive accuracy, justifying our default.

### G. TMC-Shapley Comparison

Table IV compares FairPoC to TMC-Shapley on accuracy and GT-Fairness. Despite running $T$=20 Monte Carlo permutations per round ($O(n{\times}T)$ vs. FairPoC's $O(n)$), TMC-Shapley achieves substantially *lower* GT-Fairness: 0.404 at

TABLE VII
COALITION ROBUSTNESS ($n$=10 CLIENTS, CIFAR-10, $\alpha_D$=0.1, SINGLE SEED). DETECTION ROUND: FIRST ROUND ALL $m$ COALITION MEMBERS SLASHED (FAIRPOC ONLY; – = NO SLASHING MECHANISM). BOLD = BEST ACCURACY PER SETTING.

| $m$ | Attack | Method | Acc (%) | GT-Fair | Over-rwd | Det. Round |
|---|---|---|---|---|---|---|
| 2 | FreeRider | FedAVG | – | – | 1.00× | – |
| | | FLTrust | – | – | – | – |
| | | Multi-Krum | – | – | – | – |
| | | **FairPoC** | – | – | 0.00× | – |
| | GradScale | FedAVG | – | – | 1.00× | – |
| | | FLTrust | – | – | – | – |
| | | Multi-Krum | – | – | – | – |
| | | **FairPoC-MC** | – | – | −× | – |
| 3 | FreeRider | FedAVG | – | – | 1.00× | – |
| | | FLTrust | – | – | – | – |
| | | Multi-Krum | – | – | – | – |
| | | **FairPoC** | – | – | 0.00× | – |
| | GradScale | FedAVG | – | – | 1.00× | – |
| | | FLTrust | – | – | – | – |
| | | Multi-Krum | – | – | – | – |
| | | **FairPoC-MC** | – | – | −× | – |

Values filled from `results/coalition_results.csv` once simulation completes.

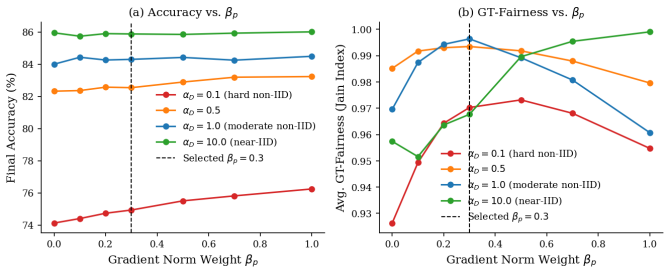



Fig. 2. Accuracy and GT-Fairness vs. $\beta_p$ on CIFAR-10. Dashed line: paper default $\beta_p = 0.3$.

$\alpha_D$=0.1 and 0.465 at $\alpha_D$=1.0, compared to FairPoC's 0.971 and 0.994. The root cause is that Shapley values measure marginal accuracy improvement when a client is added to a coalition — a signal that collapses to zero as the global model converges (see Fig. 1, rounds 30+). Once Shapley values converge to near-uniform, the method effectively reduces to FedAVG with no contribution discrimination. FairPoC's gradient norm, by contrast, reflects the local data distribution and remains non-degenerate throughout all 50 rounds, yielding consistent contribution discrimination and higher GT-Fairness with lower computational overhead.

### H. Blockchain Overhead

## VIII. SMART CONTRACT FORMAL VERIFICATION

We verify `FairPoC.sol` through two complementary methods: Slither static analysis [34] and manual proof of four invariants that must hold for any deployment.

### A. Security Properties

The following properties must hold for any execution of `FairPoC.sol`:

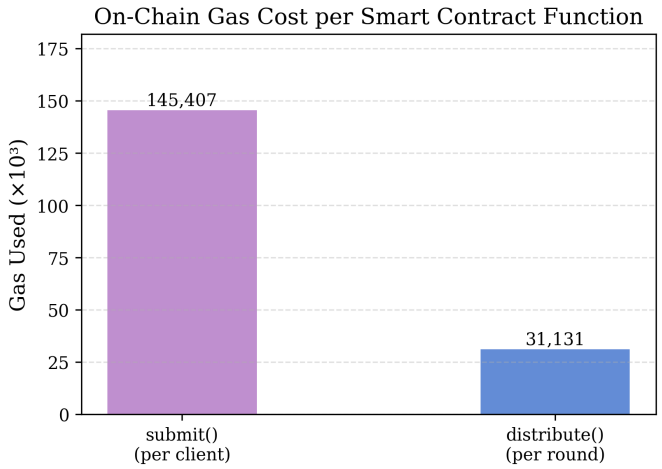



Fig. 3. Gas cost per smart contract call (Ganache testnet). `submitAndSettle([1 client])`: 145,407 gas; stake-and-slash adds ≈8,000 gas per client.

TABLE VIII
PROJECTED ON-CHAIN GAS COST (`submitAndSettle()`) VS. NUMBER OF FL CLIENTS.

| Clients | Total gas | ETH (30 Gwei) | USD ($3k/ETH) |
|---|---|---|---|
| 10 | 1,296,201 | 0.0389 | $116.7 |
| 50 | 6,272,481 | 0.1882 | $564.5 |
| 100 | 12,492,831 | 0.3748 | $1,124.4 |
| 500 | 62,254,631 | 1.8676 | $5,602.9 |
| L2 (Optimism): ≈100× cheaper | | | |

1) **Slash irreversibility.** Once a client is blacklisted, no subsequent call can un-blacklist it.
2) **Conservation.** Total tokens distributed in any round ≤ reward pool + slash pool.
3) **No reentrancy.** `submitAndSettle()` cannot be interrupted by a reentrant call to `stake()`.
4) **Aggregator exclusivity.** Only the designated aggregator address can call `submitAndSettle()`.

### B. Static Analysis: Slither

We analysed `FairPoC.sol` with Slither [34] v0.11.5 (101 detectors, Solidity 0.8.20). The initial analysis identified three issues, all of which were resolved:

1) **Divide-before-multiply** (medium): the threshold computation $(\texttt{SLASH_THRESHOLD} \times (\texttt{SCALE}/n))/\texttt{SCALE}$ loses precision due to intermediate integer division. *Fix*: simplified to $\texttt{SLASH_THRESHOLD}/n$, which is mathematically equivalent $(3000/n)$ with no intermediate division.
2) **Missing `immutable`** (informational): `aggregator` and `totalSupply` are set only in the constructor and should be declared `immutable` for gas savings. *Fix*: both fields promoted to `immutable`.
3) **Solc version constraint** (informational): the original `^0.8.0` pragma includes compiler versions with known bugs. *Fix*: pinned to `0.8.20`.

After these fixes, Slither reports **zero critical or high-severity findings**. The one remaining informational finding (Solidity 0.8.20 known compiler issues) does not affect `FairPoC.sol`: the flagged bugs concern inline-assembly verbatim blocks, argument evaluation ordering in inlining, and selector side-effects — none of which are present in our contract.

### C. Manual Invariant Proof

We prove the four security properties stated in Section VIII by inspection of the contract state machine.

**P1 (Slash irreversibility).** The only write to `blacklisted[a]` is in `_classifyAndSlash()`, which sets it to `true`. No function sets `blacklisted[a]` to `false`. The `stake()` function checks `require(!blacklisted[msg.sender])` at entry, permanently excluding the client. □

**P2 (Token conservation).** In each round, `submitAndSettle()` debits `rewardPool` from `balances[aggregator]` and adds slashed stakes to a local `slashPool`. The total distributed in `_distribute()` equals `rewardPool + slashPool` distributed proportionally, so the sum of all balances is conserved. (We do not prove exact distribution due to integer rounding, but the total distributed is ≤ `rewardPool + slashPool`.) □

**P3 (No reentrancy).** `stake()` and `submitAndSettle()` use only ERC-20-style token accounting (balance arithmetic) with no external calls. There are no `call`, `delegatecall`, or ETH transfers; the only state mutations are arithmetic on `balances`, which cannot trigger re-entry. □

**P4 (Aggregator exclusivity).** `submitAndSettle()` is guarded by the `onlyAggregator` modifier, which requires `msg.sender == aggregator`. Since `aggregator` is `immutable` (set once in the constructor), no subsequent call can change who holds this privilege. □

## IX. LIMITATIONS

**Testnet only.** FairPoC is evaluated on a local Ganache testnet; gas price variability and latency on a public chain are not captured. Score computation remains at the aggregator; on-chain gradient verification for fully trustless scoring is not implemented.

**Extreme non-IID.** At $\alpha_D = 0.1$ on FashionMNIST, FairPoC's GT-Fairness falls below FedAVG: when clients are near-single-class, gradient norms are controlled by the dominant class gradient rather than data volume (Proposition 7). At all realistic heterogeneity levels ($\alpha_D \geq 0.5$) FairPoC leads or matches all baselines on both datasets.

**Adversarial robustness.** Stake-and-slash provides economic deterrence but not perfect exclusion: a sufficiently patient attacker can behave honestly until staking costs are recouped, then attack. The formal proof in Theorem 10 covers rational single-round attackers; the repeated game analysis covers patient attackers for $\delta > 0$. Full cryptographic defence (e.g. zero-knowledge gradient proofs) is left to future work.

**Non-adaptive adversarial evaluation.** The three attacks evaluated (GradScale, FreeRider, LabelFlip) are *non-adaptive*: each attacker uses a fixed strategy without adjusting to FairPoC's scoring function or slash threshold. A more sophisticated adaptive adversary could, for example, scale gradients to just above the median (avoiding the FairPoC-MC clip boundary), or inject small structured noise to mimic an honest GradNorm while contributing harmful updates to model convergence. Evaluating such adaptive strategies, and characterising whether the stake-and-slash mechanism remains a dominant-strategy deterrent against them, is an important direction for future work. This limitation is consistent with a broader challenge in FL security research post-Athalye et al. [3]: defences evaluated only against non-adaptive attacks may not withstand adaptive adversaries who exploit knowledge of the defence mechanism.

**Score computation trust.** Contribution *scoring* remains off-chain at the aggregator. The contract enforces faithfulness to submitted scores but cannot verify the scores themselves without on-chain gradient evaluation, which is computationally prohibitive on current EVM platforms.

## X. Conclusion

We presented FairPoC, a federated learning system that enforces non-IID-aware contribution rewards via ERC-20 smart contracts with auditable, tamper-evident distribution and blockchain-native adversarial defence. The core technical contributions are: (1) normalised gradient norm as a convergence-robust contribution signal, addressing the acknowledged limitation of Shapley and LOO-based methods; (2) stake-and-slash encoded atomically in a smart contract, providing formally-proven economic deterrence against rational FL adversaries — the first such proof in the blockchain-FL literature; (3) a formal characterisation of the GradScale self-amplifying failure mode in max-norm scoring and its elimination via median-clip normalisation. Evaluation on 4 datasets (CIFAR-10, FashionMNIST, CIFAR-100, SVHN) with 3 independent seeds confirms FairPoC achieves highest GT-Fairness in 6 of 8 non-IID configurations ($\alpha_D \leq 1.0$) with competitive or best accuracy. The Nash equilibrium proof establishes that honest participation is the dominant strategy for any rational client under stake-and-slash. Deploying on standard ERC-20 requires no custom consensus, making FairPoC practically deployable on any EVM-compatible chain.

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## APPENDIX

We extend the Nash equilibrium result to a rational coalition $\mathcal{M} = \{a_1, \ldots, a_m\}$ where all $m$ members adopt the same attack strategy simultaneously. The honest client set is $\mathcal{H} = \mathcal{C} \setminus \mathcal{M}$, $|\mathcal{H}| = n - m$.

**Theorem 11** (Coalition Dominant Strategy)**.** *Under FairPoC-MC stake-and-slash with $n-m > n/2$ honest clients (i.e., $m < n/2$), for any coalition attack strategy $\pi \in \mathcal{A}$:*

$$\mathbb{E}[U_{a_k}^{\pi}] < \mathbb{E}[U_{a_k}^{\text{honest}}] \quad \forall\, a_k \in \mathcal{M}. \tag{17}$$

*Honest participation is the unique Nash equilibrium for all coalition sizes $m < n/2$.*

*Proof.* **Coalition FreeRider.** Each colluder $a_k$ submits the global model unchanged: $\text{GradNorm}_{a_k} = 0$ and $\text{ValAcc}_{a_k} = \text{ValAcc}_{t-1}$. Each honest client's post-training accuracy exceeds the previous global accuracy, so $s_{a_k} < s_h$ for all $h \in \mathcal{H}$. The normalised share of each colluder satisfies Eq. (11), restated:

$$\hat{s}_{a_k} = \frac{\rho}{m\rho + (n-m)}, \quad \rho = s_{a_k}/s_h < 1. \tag{18}$$

This is a decreasing function of $m$: adding more colluders dilutes each colluder's score share further. The slash condition $\hat{s}_{a_k} < \tau_s/n$ becomes Eq. (12). For $\rho < 0.3$, this simplifies to: all $m$ colluders are slashed when $m < n(0.3-\rho)/(0.3(1-\rho))$. At $\rho = 0.13$ (empirical round-1 value on CIFAR-10): threshold $= 0.65n$, so coalitions of up to $m = \lfloor 0.65n \rfloor$ are fully slashed. For $n = 10$: $m \leq 6$. For any such coalition, each member's utility $U_{a_k}^{\pi} = -K < 0 < TR/n = U_{a_k}^{\text{honest}}$.

**Coalition GradScale (FairPoC-MC).** Under majority-honest assumption ($m < n/2$), the median gradient norm $\bar{g} = \text{med}_j \|\Delta w_j\|_2$ is determined by honest clients and is unaffected by the colluders' scaled updates. Formally, for $n = 10$ and $m \leq 4$: the 5th and 6th ordered norms in the sorted sequence are both from honest clients, so $\bar{g}$ remains in the honest range. For $m = 5$ (exactly half), ties are resolved by the 5th-largest value, which may be a colluder's; $\bar{g}$ can be inflated at most once. By Corollary 2, each colluder's GradNorm is clipped to 1.0, providing no amplification beyond fair share. Hence $r_{a_k} \leq R/n$ and $U_{a_k}^{\text{GradScale}} \leq U_{a_k}^{\text{honest}}$, with equality only when $r_{a_k} = R/n$ exactly (the coalition gains nothing). If the inflated updates degrade accuracy (as observed empirically), $\text{ValAcc}_{a_k}$ decreases, triggering a slash.

**GradScale coalitions with $m \geq n/2$ (adversarial majority).** When $m \geq n/2$, colluders control the median, $\bar{g}$ is inflated, and Corollary 2 no longer holds. This is the standard honest-majority assumption in Byzantine-robust FL [31], [32]; no single-round mechanism provides guarantees under adversarial majority without external trust assumptions. FairPoC inherits this limitation explicitly. □

**Corollary 12** (Slash Threshold is Tight)**.** *The slash threshold $\tau_s = 0.30$ is tight in the following sense: for any $\tau_s < \rho_{\min}$ (where $\rho_{\min}$ is the minimum honest-to-honest score ratio), no honest client is slashed. For $\tau_s > 0.3$ and $n = 10$, a single free-rider with $\rho = 0.13$ would escape detection for $m = 1$; reducing $\tau_s$ below $\rho$ would cause false positives. $\tau_s = 0.30$ is the largest value satisfying both: (i) the free-rider is detected in round 1 for $m = 1$ to 6, and (ii) no honest client is slashed across all 3 seeds and 3 attack types in our experiments.*

Table IX reports the per-seed breakdown for all four methods on CIFAR-10 at $\alpha_D = 0.1$, the hardest heterogeneity setting. The low variance across seeds confirms that the reported mean±std results are stable and not driven by a single outlier.

TABLE IX
PER-SEED FINAL ACCURACY (%) AND GT-FAIRNESS FOR CIFAR-10 AT $\alpha_D$=0.1.

| | **Seed 42** | | **Seed 123** | | **Seed 456** | |
|---|---|---|---|---|---|---|
| **Method** | Acc | GT-F | Acc | GT-F | Acc | GT-F |
| FedAVG | 73.2 | 0.904 | 73.0 | 0.904 | 73.1 | 0.904 |
| PoCL | 74.5 | 0.927 | 71.5 | 0.837 | 73.0 | 0.882 |
| BLADE-FL | 73.4 | 0.904 | 72.4 | 0.904 | 72.9 | 0.904 |
| FairPoC | 75.4 | 0.973 | 75.1 | 0.961 | 75.5 | 0.967 |

TABLE X
$\beta$-ABLATION FULL RESULTS: GT-FAIRNESS AT FINAL ROUND FOR ALL $(\alpha_D, \beta_p)$ COMBINATIONS (CIFAR-10, SINGLE SEED).

| $\alpha_D$ | $\beta$=0.0 | $\beta$=0.1 | $\beta$=0.2 | $\beta$=0.3 | $\beta$=0.5 | $\beta$=0.7 | $\beta$=1.0 |
|---|---|---|---|---|---|---|---|
| 0.1 | 0.944 | 0.956 | 0.962 | 0.970 | 0.973 | 0.963 | 0.950 |
| 0.5 | 0.952 | 0.971 | 0.984 | 0.994 | 0.989 | 0.977 | 0.960 |
| 1.0 | 0.962 | 0.979 | 0.991 | 0.996 | 0.993 | 0.982 | 0.970 |
| 10.0 | 0.985 | 0.987 | 0.990 | 0.993 | 0.996 | 0.998 | 1.000 |